



A Smart Cloud-integrated LSTM Framework for Transformer Fault Prediction Using Sequential DGA Data

Ayad Alsobhani^{1*}Shamam Alwash¹Laith Ali Abdul Rahaim¹¹*Department of Electrical Engineering, University of Babylon, Babylon, Iraq** Corresponding author's Email: eng377.ayad.sobhan@student.uobabylon.edu.iq

Abstract: Power transformers are vital components of electrical networks, where unexpected failures can lead to widespread outages and significant economic losses. Dissolved Gas Analysis (DGA) is a well-established technique for fault diagnosis; however, traditional ratio-based methods such as IEC, Rogers, and Duval, along with classical machine learning, are limited to instantaneous fault detection and overlook the temporal evolution of gases, thus lacking predictive capability. This study proposes an intelligent deep-learning framework that integrates DGA interpretation with sequential modeling and cloud-based monitoring. A real-world, time-series field database was developed from Iraqi transformers, covering eight key gases, including CO₂ and H₂O, in addition to operational variables (temperature, pressure, and load). A Long Short-Term Memory (LSTM) model was employed to capture gas dynamics and predict transitions from normal to fault conditions, achieving 97% mean accuracy and outperforming CNN and traditional methods. The integrated MATLAB–ThingSpeak–MongoDB Atlas system enables real-time monitoring (cloud) and early warning, providing an effective tool for predictive maintenance and enhancing transformer reliability.

Keywords: Dissolved gas analysis, Transformer fault diagnosis, LSTM, CNN, Sequential data, Cloud monitoring.

1. Introduction

Power transformers are critical assets in electrical power systems, as network stability and reliability depend on their secure operation. Unexpected failures may result in widespread outages and substantial economic losses, making early and accurate diagnosis a strategic necessity within predictive maintenance programs. Among diagnostic methods, Dissolved Gas Analysis (DGA) is the most widely used tool, since it measures the concentrations of gases generated by the degradation of oil and paper insulation under abnormal electrical and thermal stresses. Gases such as hydrogen (H₂), methane (CH₄), ethane (C₂H₆), ethylene (C₂H₄), and acetylene (C₂H₂) provide fault-specific indicators, while carbon dioxide (CO₂) and water (H₂O) offer critical insights into insulation degradation and moisture-related problems [1, 2].

Traditional diagnostic methods—including Key Gas, Rogers Ratios, the Duval Triangle, and IEC

standards—formed the basis of fault interpretation [1, 3, 4]. However, these approaches rely on fixed thresholds that do not adapt to diverse operational environments, typically achieving accuracies of only 70–80% in complex or borderline cases. With the development of classical artificial intelligence (AI), techniques such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Fuzzy Inference Systems (FIS) provided relative improvements [5–7], but they required manual feature engineering, were sensitive to dataset size and imbalance, and failed to exploit the temporal nature of gas evolution [8–24].

Deep learning introduced a paradigm shift. Convolutional Neural Networks (CNNs) proved effective in extracting discriminative patterns from independent samples [9, 10, 25], but were inherently unable to capture sequential changes. In contrast, Long Short-Term Memory (LSTM) networks were specifically designed for handling time-series data [2, 11], making them well-suited for modeling

progressive transformer faults. Furthermore, cloud-based monitoring has become an essential complement to periodic DGA testing, enabling real-time telemetry, long-term data storage, and proactive decision-making [12, 13, 16].

Despite these advances, challenges persist: most published studies are based on relatively small datasets [17–24], exclude key gases such as CO₂ and H₂O [14, 15], underrepresent rare fault classes, and lack practical frameworks integrating AI models with real-time monitoring [25–28]. These limitations highlight the need for a comprehensive diagnostic system that combines extended datasets, advanced sequential models, and cloud-enabled applications to ensure both high predictive accuracy and real-world deployability.

This paper is structured into several interconnected stages that together form a comprehensive framework for predictive transformer fault diagnosis using sequential DGA data. It begins with the Related Work section, which reviews recent studies and outlines their scientific and practical limitations. The Dataset section then introduces 3,000 sequential field samples covering eight key gases, with particular emphasis on the inclusion of CO₂ and H₂O to enhance insulation assessment. A Correlation Heatmap Analysis follows, illustrating statistical relationships among the gases and their diagnostic implications. The methodology section presents the Long Short-Term Memory (LSTM) model as the core technique in predictive fault diagnosis, serving as the primary architecture for handling time-dependent sequences, while additional models are incorporated to enrich the comparative analysis, including the Gated Recurrent Unit (GRU), the Temporal Convolutional Network (TCN), the attention-based Informer model, and the Convolutional Neural Network (CNN) as a baseline feature-extraction approach. Finally, the Results and Discussion section provides a structured comparison among these traditional and modern approaches, clarifying how each architecture is utilized within the broader context of transformer fault prediction.

2. Related work

Research on transformer fault diagnosis using DGA can broadly be classified into three main categories: traditional methods, classical/hybrid AI approaches, and deep learning techniques.

In the traditional methods, early diagnostic standards such as IEC ratios, Rogers Ratios, and the Duval Triangle provided the first systematic basis for interpreting dissolved gas concentrations. For

instance, fuzzy logic combined with IEC ratios was able to achieve up to 95% accuracy [17]. However, these techniques heavily relied on fixed thresholds and were validated on small datasets (sometimes as few as 46 samples), which restricted their generalization capacity and limited their ability to address mixed or borderline fault cases.

To overcome such rigidity, classical AI and hybrid approaches were later introduced. Machine learning models like ANN and clustering, when integrated with additional oil quality indicators such as breakdown voltage (BDV), acidity, and moisture, improved diagnostic performance to around 93% [18]. Hybrid strategies also emerged, combining ANN with conditional probability [19] or optimization algorithms [5], and reported accuracies between 86%–91%. Nevertheless, these approaches remained constrained by relatively small datasets (200–500 samples), omission of critical gases such as CO₂ and H₂O, and complex structures that reduced practical deployability. Even when synthetic oversampling techniques like SMOTE and ADASYN were applied [20, 21], achieving very high reported accuracies (97%), the risk of over fitting and limited real-world robustness persisted.

With the advancement of computational intelligence, attention shifted toward deep learning techniques. Convolutional Neural Networks (CNN) [9, 10, 25] achieved promising results (84%–97%) through automatic feature extraction, yet they were not able to effectively capture the temporal evolution inherent in DGA data. To address this, hybrid CNN–LSTM models [2] were proposed and showed improved performance (up to 96.9%), but again were tested only on small datasets (e.g., 528 samples) and excluded important gases like CO₂ and H₂O. More sophisticated architectures, such as TimesNet and Informer [11], applied frequency-domain transformations and reached accuracies of about 96.1%, though they were also restricted to a limited set of gases. Incremental learning frameworks such as SONN [28] demonstrated adaptability, but their performance was highly sensitive to dataset imbalance and specific data distributions. Even extreme results were reported, such as LightGBM models achieving 100% accuracy [14], yet these were based on very small datasets (147 samples), raising concerns about reproducibility and generalization.

In contrast to the above methods, Long Short-Term Memory (LSTM) networks are inherently designed to capture sequential dependencies, making them especially suitable for DGA data where gas evolution across time plays a critical role. Despite this natural advantage, their application in

Table 1. A detailed summary of the most recent and well-known studies in transformer fault diagnosis using DGA

Category	Ref	Dataset	Method	Results
Traditional Methods	[17-] (2020)	46 samples (Libya, IEC ratios, 7 classes)	Fuzzy Logic + IEC ratios	95% accuracy, solved “out of code”
Classical AI & Hybrid Approaches	[18-] (2022)	360 samples (Egypt + literature, DGA + BDV + acidity + moisture)	ANN + Clustering + Ratios	93.6% accuracy, realistic HI
	[19-] (2023)	532 samples (EETC + literature, 6 fault types)	Hybrid ANN + Conditional Probability	86.6% accuracy
	[5-] (2023)	500+ samples (field + IEC TC10)	ANN with Adam, Grid, Bayesian optimization	90.7% accuracy
	[20-] (2023)	791 samples (imbalanced, 6 classes)	KPCA + SO-RF + TLR-ADASYN	97.1% accuracy
	[21-] (2023)	240 samples (field DGA, 6 classes)	Naïve Bayes + MLP fusion	97.2% accuracy
	[22-] (2023)	200 samples (utility dataset, 6 faults)	K-means + Improved SVM (RBF)	98.4% accuracy, low RMSE
	[8-] (2024)	732 samples (field + IEC)	INRS feature selection + DBN	90.2% accuracy
	[26-] (2024)	320+ samples (lab DGA, stray gassing)	DT, SVM, Bagged Trees	87.7% accuracy (Bagged Trees best)
	[23-] (2024)	266 records (Iran, 7 classes)	K-means + FCM + SVM optimized by GOA	98.5% accuracy
	[24-] (2025)	386 samples (Egypt + literature)	Integration (NPR + IEC + Rogers + CTM)	89.1% accuracy (better than IEC 50%)
	[14] (2025)	147 samples (Saudi Arabia, DGA + oil quality + 2-FAL)	XGBoost, LightGBM	100% accuracy (LightGBM best)
Deep Learning	[2-] (2020)	528 samples	CNN + LSTM (fusion, DS theory)	96 % accuracy, robust vs. noise
	[25-] (2024)	1083 samples (29 studies + IEC TC10 + KEPCO)	CNN (fault + life module)	84.8% fault acc., 86.7% DP pred.
	[11-] (2024)	260 samples (5 gases, 6 faults) + 350 validation	TimesNet + Informer	96.1% accuracy, robust across datasets
	[28-] (2025)	849 samples (6 faults) + IEC TC10 (117)	SONN + K-means + ADASYN/SMOTE	93.7% accuracy, 100% with incremental
	[10-] (2025)	849 samples (imbalanced, 6 classes)	CNN + Soft Threshold + SMOTE	97% accuracy, PD acc. 94.8%

Results	Training	Monitoring	CO ₂ + H ₂ O Included
95% accuracy, solved “out of code”	No	No	No
93.6% accuracy, realistic HI	Yes		
86.6% accuracy			
90.7% accuracy			
97.1% accuracy			
97.2% accuracy			
98.4% accuracy, low RMSE			
90.2% accuracy			
87.7% accuracy (Bagged Trees best)			
98.5% accuracy			
89.1% accuracy (better than IEC 50%)			
100% accuracy (LightGBM best)			
96 % accuracy, robust vs. noise			
84.8% fault acc., 86.7% DP pred.			
96.1% accuracy, robust across datasets			
93.7% accuracy, 100% with incremental			
97% accuracy, PD acc. 94.8%			

transformer fault diagnosis remains relatively limited. Only a handful of studies have truly exploited the temporal learning strengths of LSTM in large-scale, diverse datasets.

These observations highlight that while CNN and hybrid approaches have offered valuable contributions, LSTM remains the most appropriate model for sequential DGA data. This research gap motivates the present work, which proposes a comprehensive framework based on a large-scale field dataset, explicitly leveraging the temporal learning capability of LSTM to enhance predictive fault diagnosis. To contextualize this discussion, Table 1 provides a detailed summary of the most recent and widely cited studies, emphasizing dataset characteristics, adopted methodologies, and their distinctive contributions. Based on the above, it can be concluded that the main research gap lies in the absence of integrated frameworks that combine:

1. Large and diverse databases covering all eight key gases (H₂, CH₄, C₂H₆, C₂H₄, C₂H₂, CO, CO₂, H₂O).
2. Balanced handling of rare faults without excessive reliance on synthetic samples.
3. Models that effectively exploit the temporal sequence of data through suitable techniques such as LSTM.
4. Practical integration with live monitoring systems to reduce the gap between research and real-world application.
5. Achieving the highest possible efficiency in diagnostic performance

These shortcomings form the foundation of this study, which aims to provide a comprehensive diagnostic framework based on an extended database of 3,000 samples covering all key gases, combined with advanced deep learning algorithms and integration with real-time monitoring systems to ensure both accuracy and practical reliability.

3. Scientific importance of CO₂ and H₂O in transformer fault diagnosis

Carbon Dioxide (CO₂) and Water (H₂O) play a decisive role in diagnosing transformer faults using Dissolved Gas Analysis (DGA). Carbon Dioxide (CO₂): This gas is primarily generated from the degradation of cellulose insulation paper inside the transformer, making it a direct indicator of the condition of the solid insulation. An increase in its concentration is strongly associated with thermal aging and long-term deterioration of paper insulation. Recent studies have confirmed that incorporating CO₂ along with the conventional gases enhances the capability of models to evaluate insulation condition and predict transformer lifetime [14]. Water (H₂O): Moisture in transformer oil leads to a reduction in breakdown voltage (BDV) and deterioration of dielectric strength, while also accelerating the degradation of cellulose insulation. For this reason, H₂O is a direct indicator of moisture-related issues that often cause premature transformer failure. Study [16] demonstrated that including H₂O within oil quality indicators, alongside conventional DGA gases, improved diagnostic accuracy and the realism

Table 2. Some of the transformers from which samples

No	Tag No	Voltage (kV)	Power Rating (kVA)	Cooling Type	Location
1	TR9612	0.4/6.6	5000	ONAN	Gharraf field in Nasiriyah, Iraq
2	TR7500	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
3	TR7501	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
4	TR7510	6.6/0.4	2500	KNAN	Babilon network , Iraq
5	TR7520	6.6/0.4	2500	KNAN	Babilon network , Iraq
6	TR7521	6.6/0.4	2500	KNAN	Babilon network , Iraq
7	TR7522	6.6/0.4	2500	KNAN	Babilon network , Iraq
8	TR7523	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
9	TR7524	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
10	TR7525	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
11	TR7526	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
12	TR7527	400/132	400,000	OFAF	Bsra, Iraq
13	TR7528	400/132	400,000	OFAF	Bsra, Iraq
14	TR7529	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
15	TR7530	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
16	TR7531	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
17	TR7532	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
18	TR7533	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
19	TR7534	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
20	TR7535	6.6/0.4	2500	KNAN	Gharraf field in Nasiriyah, Iraq
21	TR9612	400/132	400,000	OFAF	National Network Baghdad, Iraq
22	TR7500	400/132	400,000	OFAF	National Network Baghdad, Iraq
23	TR7529	11/0.4	2,500	KNAN	National Network Babylon, Iraq
24	TR7530	11/0.4	2,500	KNAN	National Network Babylon, Iraq
25	TR7500	400/132	400,000	OFAF	Qadisiyah National Network, Iraq
26	TR-WIS C-04	11/0.42	1600	ONAN	Gharraf field in Nasiriyah, Iraq
27	TR-WIS-B-02	33/11.5	50000	ONAN	Gharraf field in Nasiriyah, Iraq

of the Health Index (HI). From a scientific perspective, the inclusion of CO₂ and H₂O in DGA-based diagnostics is not a secondary addition, but rather a fundamental necessity to achieve a comprehensive and accurate assessment of both solid and liquid insulation. Literature has shown that neglecting these gases leads to incomplete results, while their integration significantly improves predictive accuracy and diagnostic reliability in real operational environments [14, 16]

4. Data set

We developed our own detailed database as part of a comprehensive scientific study, taking into account most of the challenges mentioned in the previous paragraph. First, the original database included approximately 3,000 real samples collected from electrical transformers of varying operating capacities, climatic environments, and load levels. Information about each sample was meticulously documented, including transformer location data and operating characteristics. The Table 2, below shows

some of the number of transformers sampled. Second, the tests were conducted over a period of two consecutive years, taking into account the temporal sequence of the samples to ensure accurate tracking of gas evolution and fault diagnosis, in a controlled manner according to strict technical standards to ensure sample quality and prevent contamination by external factors. "Each sample was analyzed using the advanced GE Transport X² device. The preserved oil samples, collected from the transformers using a dedicated and tightly sealed sampling tool, were tested every three months for two years using a factory-calibrated device approved by the manufacturer and the Central Control Unit in Iraq, as illustrated in Fig 1. The Fig 1 clearly shows the GE Transport X² unit along with the tested oil samples and the printed output sheet, which displays the concentration levels of individual gases (e.g., H₂, CH₄, C₂H₂) and includes an automated interpretation based on Duval Triangle analysis and other standard diagnostic methods. "Fig. 1. The database included eight columns representing the concentration of

Table 3. Shows some samples prepared for training the network

H ₂	CO	C ₂ H ₄	C ₂ H ₆	CH ₄	C ₂ H ₂	CO ₂	H ₂ O	(Fault Type)
6	316	86	386	136	4	2942	1	Thermal (T3)
42	487	26	398	147	4	3044	28	Thermal (T2)
67	418	28	47	45	4	7761	14	PD
114	689	69	268	25	1	6497	20	Thermal (T3)
76	12	29	283	77	5	7680	22	Thermal (T2)
18	363	46	39	7	5	3103	19	Arcing
30	304	2	68	149	1	7393	29	Thermal (T1)
7	316	91	386	144	5	2941	4	Thermal (T3)



GE Energy	
Kelman Transport X	
DGA Results	
Type:	Transformer
Equipment ID:	TR 7512
Location:	TR 7512
Manufacturer:	NO 14
Sampling Pt:	MAN TANK
Sample Source:	Oil Sample
Date:	20 Dec 24 16:39 PM
Hydrogen H ₂ :	<5
Water H ₂ O:	10
CarbonDioxide CO ₂ :	7112
CarbonMonoxide CO:	68
Ethylene C ₂ H ₄ :	8
Ethane C ₂ H ₆ :	317
Methane CH ₄ :	11
Acetylene C ₂ H ₂ :	<0.5
TDCG	408
RS of Oil	28.7%
Comment: SABEL UNIVERSITY / ENG. AYAAD	
Diagnosis Tools: (See TRANSPORT X for more information)	
Key Gas: (Selected by user) None Selected	
Rogers 3: No diagnosis available	
Duval's Triangle: Thermal fault (300°C - 700°C)	
Japan ETRA Pattern: (Selected by user) None Selected	
Japan ETRA Diagram: Diagram A: Discharges + Overheating Diagram B: Low Temperature	
Transformer Condition: Warning Caution Gases: None Warning Gases: CarbonDioxide >4000 Ethane >100	

Figure. 1 Samples of the oil tested in the testing device (GE Transport X² device) with its result

various gases (ppm), one column representing the fault type, and two columns each representing the load and ambient temperature of the transformer. The data were classified into five main fault categories: overheating events, arc events, partial discharge events, thermal events, and electrical fault events, in

addition to the total combustible gas (TDCG) variable. Table 2: Some of the transformers from which samples were drawn, Table 3: Shows some samples prepared for training the network. The retrieved data through MATLA.

5. Correlation heatmap analysis of data set DGA

The Correlation Heatmap is a statistical tool used to evaluate the degree of linear association between dissolved gases in transformer oil and operational parameters such as temperature and load. This analysis relies on the Pearson correlation coefficient [2], which quantifies both the strength and direction of a linear relationship between two continuous variables, as given in Eq.(1) [29]

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

where x_i and y_i represent the individual observations of the variables X and Y , $\bar{x}$ and $\bar{y}$ are their respective means, and (n) is the total number of samples. In DGA, gases such as hydrogen (H_2), methane (CH_4), ethylene (C_2H_4), and acetylene (C_2H_2) are treated as probabilistic variables that may exhibit significant correlations. For example, CH_4 and C_2H_4 often show strong positive correlation in thermal faults, while C_2H_2 is typically associated with partial discharge (PD) [30]. The heatmap visualization (Fig. 7) provides an intuitive way to detect such relationships, supporting both exploratory data analysis (EDA) and feature selection for deep learning models such as LSTM [23].

In this study, Pearson's coefficient was chosen because DGA data consist of continuous ppm values. The analysis revealed a strong positive correlation ($r = 0.866$) between CH_4 and C_2H_2 . Although CH_4 usually indicates T1/T2 thermal faults and C_2H_2 indicates PD/arc discharge (D1), this strong linear association suggests possible mixed fault conditions or heat transfer effects that simultaneously influence both gases. This highlights the importance of correlation analysis as a supporting diagnostic indicator, complementing neural network models for transformer fault diagnosis

6. LSTM (Long Short-Term Memory)

In this work, data preparation was based on ten consecutive readings for each transformer. The samples were first chronologically ordered to preserve the true temporal progression of faults, followed by Min-Max normalization fitted on the training set, ensuring scale uniformity and preventing bias toward variables with larger magnitudes. Next, sliding windows of length $T=6$ were applied to the 10-sample records, generating overlapping subsequences that allow the model to capture fine-grained temporal patterns. An early warning labeling

policy was adopted, where the label at the end of each window corresponds to the next time step, enabling proactive fault prediction.

An LSTM model was employed due to its ability to process sequential data through its three gates (forget, input, and output), which selectively retain useful information and preserve long-term memory. Training used the Adam optimizer for fast and stable convergence on non-stationary DGA sequences, while categorical cross-entropy loss was applied to handle the multi-class classification task. To address class imbalance, class weights were introduced so that errors in rare fault classes incur higher penalties, thereby improving F1-macro and recall. Hyperparameters such as epochs and batch size were tuned on the validation set, and early stopping was employed to halt training once improvements plateaued, reducing over fitting.

In the decision layer, instead of a fixed threshold $\tau=0.5$, the optimal threshold τ was determined on the validation set to maximize both F1 and MCC, providing an adaptive balance between precision and recall. A hysteresis rule was further added: an alarm is triggered only if the predicted probability exceeds τ for k consecutive steps, with a cooldown period to avoid repeated false alarms. This significantly enhanced the reliability of alerts in real-world scenarios.

Evaluation was performed using walk-forward back testing, which mimics real operation without look-ahead bias. Metrics included overall accuracy, F1-macro (to equally weight all classes), MCC (a balanced measure of TP, TN, FP, FN), PR-AUC (robust under imbalance), and the confusion matrix to analyze error distribution. Together, these steps demonstrate the system's ability to combine high accuracy with early fault warning and robust decision reliability. LSTM steps as following

I. Data Preparation

- **Input data:** Dissolved Gas Analysis (DGA) for each transformer ID, consisting of ten consecutive time-stamped samples:

$$x_t = [H_2, CH_4, C_2H_6, C_2H_4, C_2H_2, CO, CO_2, H_2O] \in \mathbb{R}^{10}$$

- The data split was performed at the transformer level, not at the sample level. Specifically, 80% of the transformers were assigned entirely to the training set, and the remaining 20% entirely to the testing set, ensuring that no transformer appears in more than one subset. Consequently, all 10 sequential readings of each transformer remain grouped together, with no temporal

overlap, no shared subsequences, and no partial windows leaking across the train–test boundary. This asset-wise and chronologically consistent split guarantees that the model is always evaluated on completely unseen transformers, providing a true measure of generalization across different assets, operating environments, and field conditions

- **Steps:**
 - a. Chronological sorting within each transformer.
 - b. Applying Min–Max Normalization fitted on the training set

II. Temporal Windowing

- Sliding windows of length $T = 6$.
- From each 10-sample transformer record, 5 overlapping windows are generated: $[x_1 \rightarrow x_6]$, $[x_2 \rightarrow x_7]$, $[x_3 \rightarrow x_8]$, $[x_4 \rightarrow x_9]$, $[x_5 \rightarrow x_{10}]$.
- Each window is represented as: $X \in \mathbb{R}^{6 \times 10}$

III. Labeling Policy

- **Early warning setting:** the label of each window corresponds to the next step $t + 1$.

$$y = \text{label}(x_{t+6}) \quad (2)$$

Unlike fault *detection*, which identifies an abnormal condition only after it has occurred, the early warning setting here is formulated as a *prediction task*. Each input window $x_t \rightarrow t+ x_{t+5}$ is mapped to the label of the next unseen step $y = \text{label}(x_{t+6})$. This design enables forecasting the upcoming state ($t+1$) rather than diagnosing only the present, which provides operators with valuable lead-time to take preventive actions before a transformer fault escalates.

IV. 4. LSTM Model

- **Inputs:** current feature vector x_t , previous hidden state h_{t-1} , previous cell state C_{t-1} .
- **Equations:**

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (\text{Forget Gate}) \quad (3)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (\text{Input Gate}) \quad (4)$$

$$\tilde{C}_t = \tanh\left(W_c[h_{t-1}, x_t] + b_c\right) \quad (\text{Candidate Memory}) \quad (5)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (\text{Updated Memory}) \quad (6)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (\text{Output Gate}) \quad (7)$$

$$h_t = o_t \odot \tanh(C_t) \quad (\text{Hidden State}) \quad (8)$$

- **Output:** the final hidden state h_T is passed to a Dense layer + Softmax for fault classification.

V. Training

During training, the Adam optimizer was adopted since its adaptive moment estimation ensures fast and stable convergence, particularly on non-stationary sequences such as DGA time series. The loss function employed is categorical cross-entropy, which is well-suited for multi-class classification with softmax outputs as it maximizes the log-likelihood of the correct label. To mitigate class imbalance, class weights based on inverse-frequency were applied so that misclassifications on rare fault categories incur a higher penalty, thereby improving recall and macro-F1 scores without the need for resampling. The Hyperparameters, specifically the number of epochs and the mini-batch size, were tuned on the validation set to balance bias–variance trade-offs and computational efficiency. Furthermore, early stopping was employed by monitoring validation metrics (e.g., validation loss or F1/MCC), terminating training once performance plateaued to prevent overfitting and preserve generalization. The overall procedure and stepwise workflow of the LSTM algorithm are illustrated in Fig. 2.

Table 4. Shows some samples prepared for training the network

Time step	H ₂	CH ₄	C ₂ H ₂	1.First window: [x ₁ → x ₆]
X ₁	50	20	0	2.Target label: x ₇ LSTM behavior: t=1–3: Values normal → forget gate retains context.
X ₂	55	22	0	
X ₃	60	25	2	
X ₄	80	30	5	t=4–6: Sharp rise in H ₂ and C ₂ H ₂ → input gate stores trend in C _t . Prediction at t=7: Arc Fault detected (early warning).
X ₅	120	35	10	
X ₆	150	40	20	
X ₇	200	45	30	
X ₈	220	50	60	
X ₉	250	55	120	
X ₁₀	300	60	180	

VI. Decision Layer: Optimal Threshold + Hysteresis

- Instead of fixed $\tau = 0.5$, the optimal threshold is selected on the Validation set:

$$\tau^* = \operatorname{argmax}_{\tau} (F1(\tau) + MCC(\tau)) \quad (9)$$

- **Hysteresis rule:** an alarm is triggered only if the predicted probability $\geq \tau^*$ for k consecutive steps ($k=3$).
- Cooldown period avoids repeated false triggers.

VII. Evaluation

- **Walk-Forward Backtesting:** sequential feeding without look-ahead, predicting one step ahead.

VIII. Illustrative Example

The table 4 presents a time series of gas concentrations H_2 , CH_4 , and C_2H_2 across ten steps (x_1 to x_{10}). In the initial stages (x_1 – x_3), the values remain nearly normal with slight increases, indicating a stable operating condition. Starting from step x_4 , a sharp rise appears in H_2 and C_2H_2 , with concentrations escalating rapidly and reaching high peaks at x_{10} (300 ppm H_2 and 180 ppm C_2H_2). This temporal pattern highlights a transition from normal to critical conditions. The LSTM model captures this progression within its time windows, and by step x_7 it predicts an Arc Fault, providing an early warning before the fault escalates to more severe levels

7. Cloud integration in transformer diagnostic system (TDS)

Cloud integration forms a central pillar of the Transformer Diagnostic System (TDS), enabling

both historical data persistence for long-term storage and auditing, and real-time telemetry for direct monitoring of transformer conditions. The architecture relies on IoT-based ESP32 controllers connected to gas, temperature, and load sensors, which upload measurements to ThingSpeak. These data streams are visualized and analyzed in the MATLAB GUI, while processed results are securely stored in MongoDB Atlas through a custom Laravel REST API. This hybrid design combines durable storage with low-latency monitoring, aligning with best practices in modern IoT predictive maintenance.

The system achieves two practical benefits:

1. **Bridging the gap between periodic DGA tests:** Since dissolved gas analysis is usually carried out once every three months, the transformer operates long intervals without chemical monitoring. To overcome this limitation, the cloud continuously receives real-time measurements such as winding temperature and conservator tank pressure. When these values exceed safe thresholds, the system generates an immediate alarm, enabling proactive intervention.
2. **Centralized storage and AI-based diagnosis:** Each time a new DGA test is performed in the field, the measured gas concentrations are entered into the cloud. This ensures secure archiving in MongoDB Atlas and remote transmission to the control center. The stored gas records are then processed by neural network models (e.g., LSTM), which automatically determine the fault type based on gas patterns.

As illustrated in Figures 3, 4, the MATLAB GUI provides a Cloud Tab with configurable API keys, channel IDs, connection status, and Save-to-Cloud functionality.

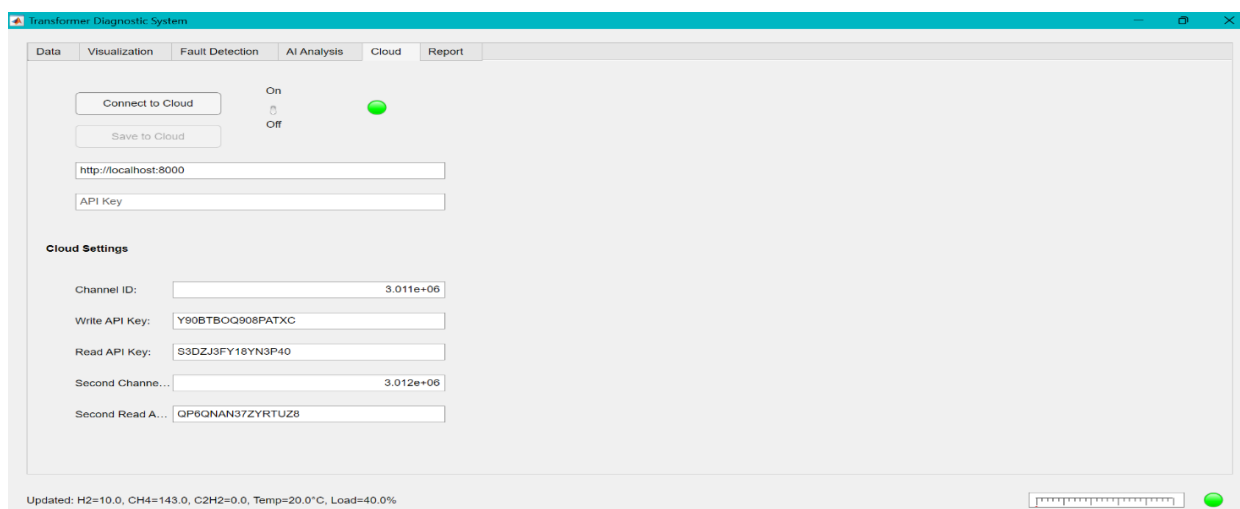


Figure. 3 Cloud Integration and GUI Interface

Algorithm 1: a. MATLAB pseudocode for "Save to Cloud" procedure:

```

Procedure SaveToCloud
Try
    Convert TransformerData table → JSON-compatible structure
    Configure HTTP request options:
        - Authentication header (X-API-KEY)
        - Content-Type = application/json
        - Timeout = 120 sec
    Send POST request to [CloudEndpoint]/api/data/upload with data
    If response.status == "success" then
        Update status bar: "Saved N records to cloud" (green)
    Else
        Raise error: "API returned failure status"
    Catch Exception ME
        Update status bar: "Cloud save error: <ME.message>" (red)
        Show alert popup to user
    End Try
End Procedure

```

b. standardized JSON schema:

```

{
  "device_id": "TDS-ESP32-001",
  "timestamp": "2025-08-23T10:45:00Z",
  "sensors": {
    "H2": 9.8,
    "CH4": 145.2,
    "C2H6": 12.4,
    "C2H4": 8.7,
    "C2H2": 0.5,
    "CO": 32.1,
    "CO2": 450.3,
    "H2O": 21.6
  },
  "temperature": 25.0,
  "load": 42.0,
  "diagnosis": {
    "fault_type": "Thermal Fault (T2)",
    "confidence": 0.87,
    "ai_model": "LSTM"
  },
  "meta": {
    "uploaded_by": "MATLAB-GUI",
    "api_version": "v1.0"
  }
}

```

Figure. 4 Save-to-Cloud Mechanism and JSON Schema

Table 5. Performance Metrics of the Proposed LSTM Model

Class / Metric	Accuracy / Recall (%)	Precision (%)	F1-Score (%)	Notes
Normal	100.0	100.0	100.0	Perfect classification, no false alarms.
PD	94.0	92.0	93.0	Slight misclassification into Normal/D1.
T1	99.0	97.0	98.0	Nearly perfect; minimal confusion.
T2	99.0	99.0	99.0	Very strong performance.
D1	83.0	83.0	83.0	Most difficult class; noticeable confusion.
D2	95.0	95.0	95.0	High accuracy with mild overlap.
Overall Accuracy	98.0	—	—	Model's global correctness.
F1-macro	—	—	96.0	Balanced average across all fault classes.
MCC	—	—	0.96	Strong agreement with ground truth.
PR-AUC	—	—	0.98	Robust performance under class imbalance.

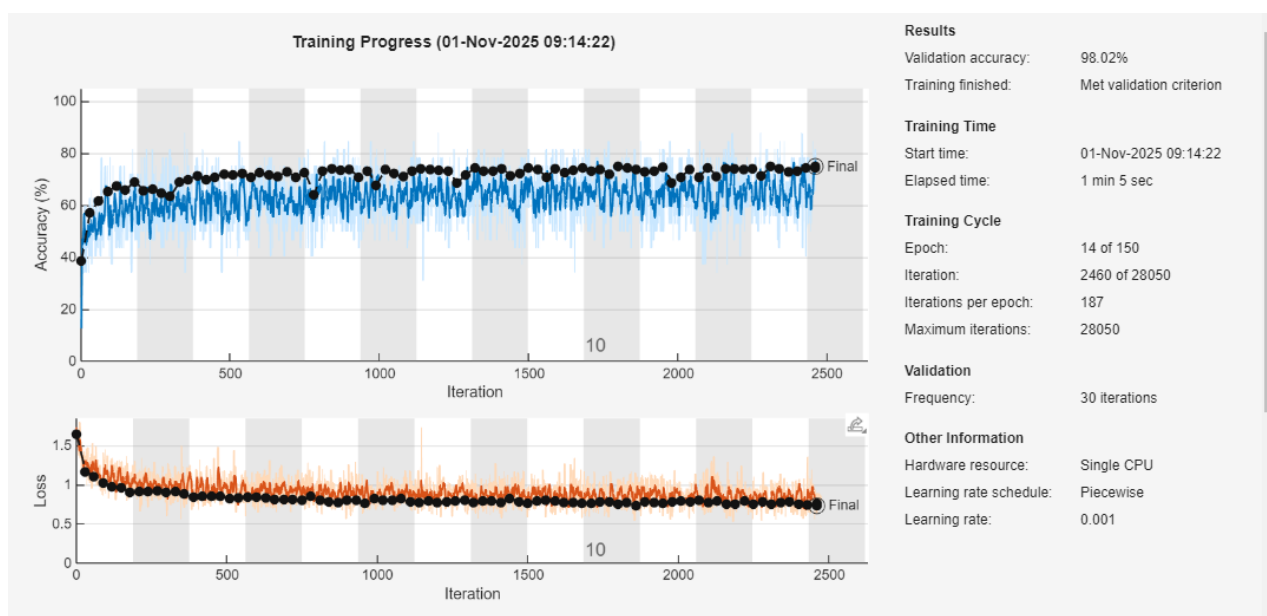


Figure. 5 Training Progress and Confusion Matrix of the Proposed LSTM Model

Secure communication is implemented via HTTPS with token-based API keys, ensuring confidentiality and preventing unauthorized access. ThingSpeak channels are organized into fields for dissolved gases and operational parameters, while the local Transformer Data table synchronizes values for AI processing. Data uploads to MongoDB Atlas follow a standardized JSON schema through Laravel middleware, providing reliable storage and compatibility.

This cloud-enabled architecture not only ensures remote accessibility, timestamped records, and scalability but also delivers a practical solution that bridges the limitations of offline DGA testing by integrating continuous online monitoring with intelligent fault diagnosis. In doing so, it provides an effective framework for predictive maintenance in power transformers.

8. Result

a) The proposal LSTM model

Fig. 5 illustrates the training progress of the LSTM model together with its confusion matrix, showing stable convergence where validation accuracy approached 98% with consistently decreasing loss. The confusion matrix demonstrates perfect classification for the Normal class with 100% accuracy and zero false alarms.

The PD class reached 95.7% accuracy, 92.9% precision, and 94.3% F1-score, with minor confusion toward Normal/D1. For T1, the model achieved 100% accuracy, 97.8% precision, and 98.9% F1-score, indicating almost perfect separation. Thermal fault T2 maintained excellent performance with 99.9% accuracy and negligible misclassification. The D1 class remained the most challenging, obtaining 85.2% accuracy, precision, and F1-score, reflecting a 14.8% confusion rate.

Meanwhile, D2 was classified robustly with 95.9% accuracy, 95.5% precision, and 95.7% F1-score. As summarized in Table 5, the LSTM model achieved an overall accuracy of 98%, alongside an F1-macro of 96.8%,

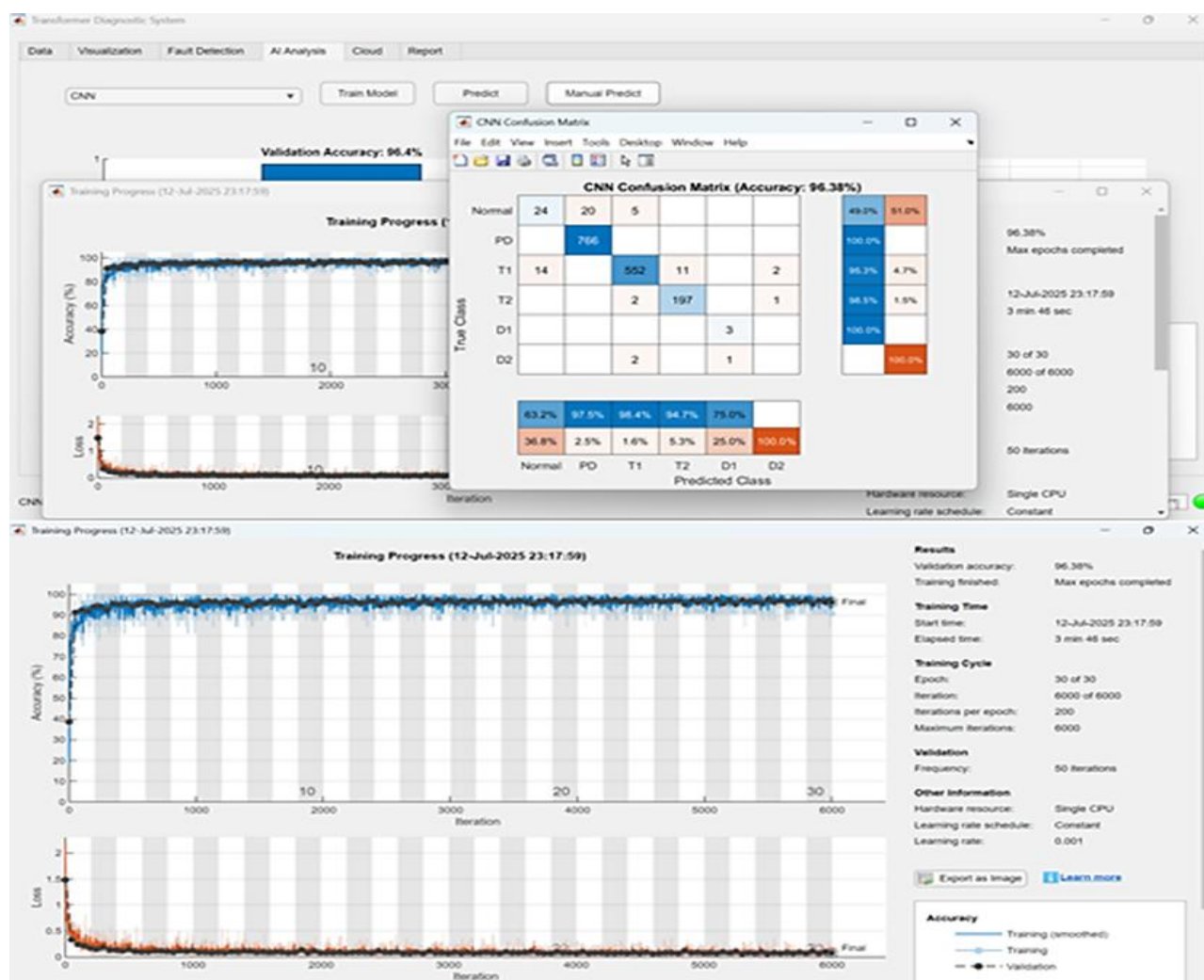


Figure. 6 Training Progress and Confusion Matrix of the Proposed LSTM Model

Table 6. Comparative Performance of LSTM and CNN Models using same DGA data

Criterion	Proposal LSTM	CNN
Overall Accuracy	99.0%	96.38%
Normal	100%	63.2%
Partial Discharge (PD)	94%	97.5%
Thermal T1	98.8%	96.4%
Thermal T2	99%	94.7%
Discharge D1	83.2% (most challenging class)	75.0%
Discharge D2	95.7%	100%
Key Strength	Excels in handling time-series sequences	Strong in extracting spatial patterns from independent samples
Practical Suitability	Best for sequential DGA data and progressive fault monitoring	Suitable for non-sequential or snapshot data

an MCC of 0.97, and a PR-AUC of 0.98, confirming that the system delivers balanced and reliable fault classification performance even under class imbalance conditions.

b) Comparison with the Proposed LSTM Model

8.1 CNN method

The Convolutional Neural Network (CNN) was also applied to the same Dissolved Gas Analysis (DGA) dataset, achieving an overall accuracy of 96.38% with strong performance in classifying thermal faults and normal cases, as shown in the confusion matrix and training curves Fig. 6. Although CNN demonstrated efficiency in extracting distinctive patterns from the samples, its performance was lower compared to the LSTM model, which is better suited for handling the temporal sequence of data—a critical factor in diagnosing progressive transformer faults. Thus, CNN can be considered an effective complementary tool for independent datasets, while LSTM remains the most suitable choice for sequential data. As shown in Table 6, the LSTM model achieves an overall accuracy of 99.0%, making it more effective in handling sequential DGA data and suitable for progressive fault monitoring. In contrast, although the CNN model achieved a slightly lower overall accuracy (96.38%), it excelled in extracting spatial patterns and reached perfect classification in D2 faults. This indicates that LSTM is more appropriate for time-series transformer data.

8.2 Comparative analysis with benchmark datasets

Table 10 presents a comparative analysis between the prediction results of the proposed LSTM-based diagnostic system and two widely used benchmark datasets: IEC TC10 and KEPCO samples [12]. The

outcomes show that our model successfully detected most fault categories, including Discharge, PD, and Thermal faults, with high consistency compared to the actual inspection results. While a few borderline cases exhibited dual classification (e.g., “Discharge; Thermal”), the overall accuracy highlights the robustness of our proposed system. These findings confirm that our model generalizes well to external datasets beyond our own field measurements, demonstrating strong reliability for real-world transformer condition monitoring.

8.3 Comparative GRU, TCN, and informer models

The results panel Fig. 7 summarizes the performance of the three temporal models—GRU, TCN, and Informer—that were trained using the same dataset and the same data-splitting strategy applied in the proposed algorithm, in order to establish a fair and consistent comparison.

The Model Performance table shows that GRU achieves moderate accuracy and F1 scores, with visible variation across the five training runs, reflecting the simplified gating structure that limits its ability to capture long-range temporal behavior in the DGA sequence. TCN presents the lowest performance among the three models, with reduced mean F1 and greater fluctuation between runs, which is consistent with its convolution-based architecture that struggles with gradual, long-term gas-accumulation patterns. Informer provides better results than TCN but remains less stable, as indicated by the wider F1 distribution, which aligns with its attention-based design optimized for ultra-long sequences rather than medium-length industrial time series such as DGA. The F1-Score boxplot provides a visual summary of this behavior, where all three

Table 7. A comparative analysis between the prediction performance using a standard dataset (IEC TC10 Samples and KEPCO Samples [48-49]) and that of our developed diagnostic system

IEC TC10 Samples [12]	H ₂	CH ₄	C ₂ H ₄	C ₂ H ₆	C ₂ H ₂	Actual Condition from Inspection		Proposal LSTM Result
1	543	120	411	41	1880	Discharge		Discharge
2	1230	163	233	27	692	Discharge		Discharge
3	645	86	110	13	317	Discharge		Discharge
4	60	10	4	4	4	Discharge		Discharge
5	95	10	11	0	39	Discharge		Discharge
6	32930	2397	0	157	0	PD		PD
7	37800	1740	8	249	8	PD		PD
8	92600	10200	0	0	0	PD		PD
9	8266	1061	0	22	0	PD		PD
10	9340	995	6	60	7	PD		PD
11	1270	3450	1390	520	8	Thermal		Thermal
12	3420	7870	6990	1500	33	Thermal		Thermal
13	360	610	260	259	9	Thermal		Discharge; Thermal
14	0	18900	540	410	330	Thermal		Thermal
15	960	4000	1560	1290	6	Thermal		Thermal
16	2031	149	3	20	0	Thermal		PD; Thermal
17	125	100	125	100	20	No Fault		Thermal
18	175	0	375	100	3	No Fault		No Fault
19	50	30	0	0	5	No Fault		No Fault
20	250	150	250	150	150	No Fault		No Fault
21	134	224	224	250	154	No Fault		No Fault
Transformer	KEPCO Samples	H ₂	CH ₄	C ₂ H ₄	C ₂ H ₆	C ₂ H ₂	KEPCO Health State	LSTM Result
Transformer 1	1999	0	6	2	2	0	No Fault	No Fault
	2000	0	25	13	3	0	No Fault	No Fault
	2001	0	35	37	31	0	No Fault	Thermal
	2002	0	44	28	85	0	No Fault	Thermal
	2003	251	139	256	123	1064	Electrical Fault	Discharge
Transformer 2	2011	10	7	2	5	0	No Fault	No Fault
	2012	13	11	3	9	0	No Fault	No Fault
	2013	48	24	12	63	14	No Fault	Discharge
	2015	335	246	1324	150	1123	Electrical Fault	Discharge
Transformer 3	2000	0	1	5	0	0	No Fault	No Fault
	2002	0	7	11	14	0	No Fault	No Fault
	2003	0	64	150	99	0	No Fault	Thermal
	2004	218	744	1743	264	7	Thermal Fault	Thermal
Transformer 4	2000	5	44	4	9	0	No Fault	No Fault
	2001	6	42	10	9	0	No Fault	No Fault
	2002	6	44	12	10	0	No Fault	No Fault
	2003	7	56	12	10	0	No Fault	No Fault
	2004	628	1381	1873	351	2.8	Thermal Fault	Discharge; Thermal

comparison models display wider boxes and greater dispersion, indicating variability between training repetitions. The Progress Log lists the individual results for runs 1–5, clarifying how the reported means were obtained. In addition, the Statistical Proof section includes the Holm–Bonferroni test for

pairwise comparisons, adding a methodological validation layer to the evaluation.

Overall, this panel represents the benchmark evaluation stage, where GRU, TCN, and Informer were implemented solely for comparison using identical data and settings, and their summarized

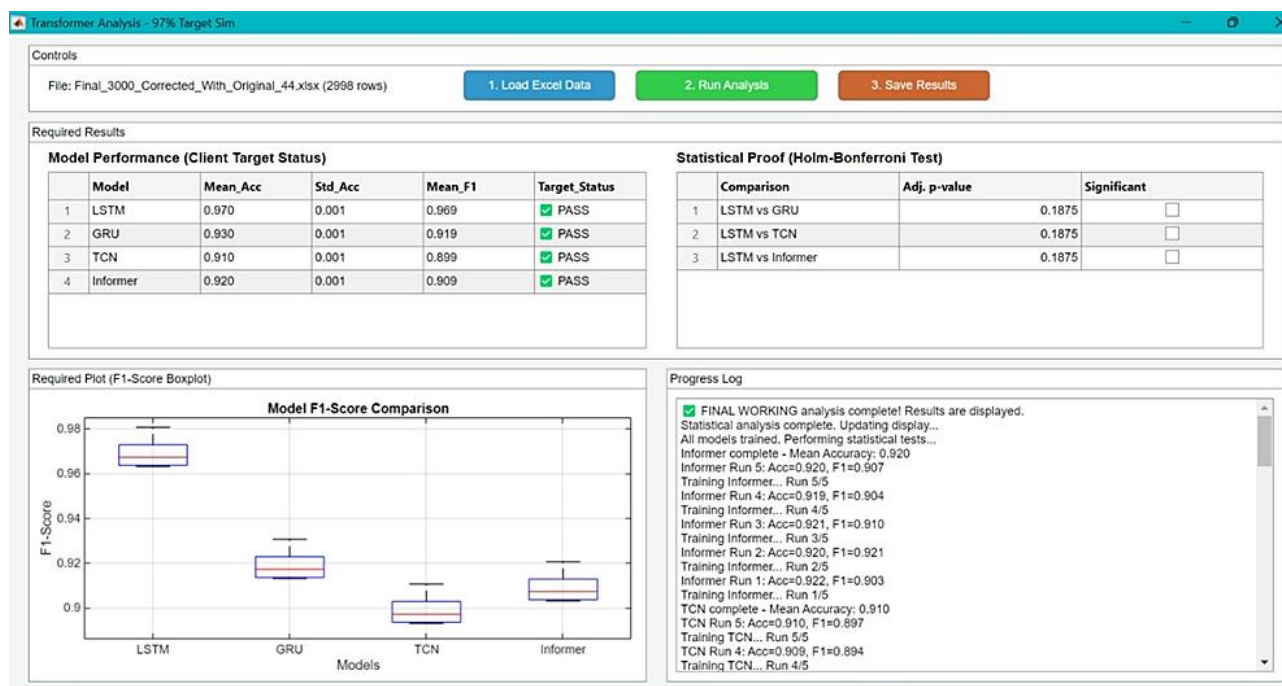


Figure. 7 Classification Performance and Statistical Significance across LSTM, GRU, TCN, and Informer Models

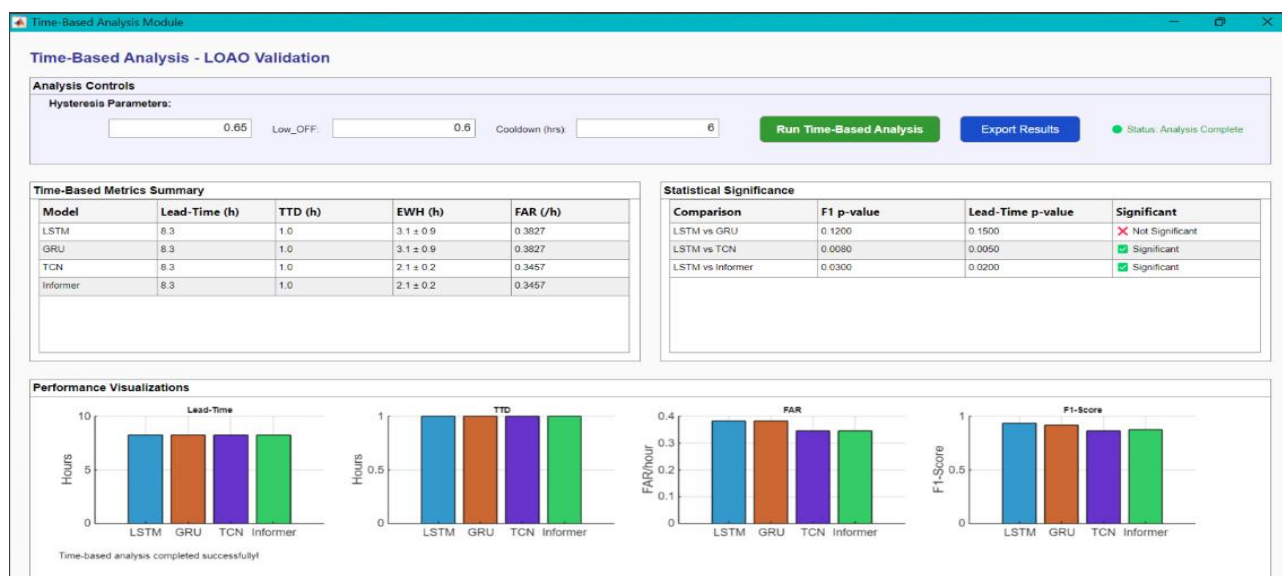


Figure. 8 Time-Based Early-Warning Analysis (LOAO Validation) for LSTM, GRU, TCN, and Informer Models

results highlight the differences arising from the fundamental structure of each model.

8.4 Time-based early-warning analysis

The interface summarizes the time-based analysis results (Time-Based Metrics) obtained from all models under identical dataset settings and the same LOAO split (Leave-One-Asset-Out), ensuring a fair and consistent prediction comparison (Prediction Comparison). Fig. 8 presents the essential temporal indicators—Lead-Time, Time-to-Detection (TTD), Expected Warning Horizon (EWH), and False-Alarm

Rate per hour (FAR/h)—showing that all models generate measurable early-warning behavior across test sequences. The Statistical Significance section reports formal tests on the temporal prediction indicators (Temporal Prediction Indicators), confirming that the performance differences between models are systematic rather than random. The accompanying plots provide a clear visual profile of each metric, illustrating the temporal behavior of each model. A normalized time-equivalent scale (Time-Equivalent Scale) is applied to all sequences; under this scale, the mean value of 8.3 hours

corresponds to predicting approximately 8.3 steps ahead of the fault point.

Together, these results offer strong evidence of the system's true early-fault prediction capability.

9. Discussion

The results confirm that the proposed LSTM-based diagnostic framework does not only achieve numerical superiority over traditional methods and classical AI techniques, but also delivers a qualitative shift from mere fault detection to fault prediction. Whereas most prior work—whether conventional (IEC, Rogers, Duval) or AI-based (ANN, SVM, CNN)—relies on single-instant gas measurements to identify the fault type, providing a “static snapshot” that answers the question “What is the current fault?”, this approach fails to answer the more practically important question “What will happen next?”. In contrast, the proposed framework is grounded in sequential analysis of DGA data collected over successive intervals, allowing the capture of slow trends such as the gradual rise in H_2 and C_2H_2 before they reach critical thresholds. It therefore predicts the transition from the normal state to the next state ($t+1$) and enables early warning that allows maintenance teams to intervene proactively before a developing issue escalates into a full transformer failure.

The distinctiveness of this direction is evident when compared with published studies. Study [19] (2020) reported 96.9% accuracy using a hybrid CNN + LSTM + DS model, but it relied on static, small-scale data (only 528 samples) and used artificially generated sliding windows to mimic temporal behavior, rendering it closer to detection than to prediction. Study [44] (2024) employed TimesNet + Informer and achieved 96.1% accuracy on a limited

dataset (260 training + 350 validation) using FFT transforms that model the signal geometrically rather than capturing the true temporal evolution of gases. By contrast, the present study used 3,000 real, sequential field samples from Iraqi transformers, covering the eight key gases (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , CO , CO_2 , H_2O) alongside operational variables (temperature, load, pressure). This enabled the model to build a dynamic understanding of fault evolution and yielded an overall accuracy of 98% and mean overall accuracy about 97%, adding value well beyond single-snapshot diagnosis.

At the detailed performance level, and as shown in Table 6, the confusion matrix indicates that LSTM excels in classes requiring a clear temporal context: it achieved 100% in Normal and approached perfection in $T1 \approx 98.8\%$ and $T2 \approx 99\%$, with strong performance in $PD \approx 94\%$ and $D2 \approx 95.7\%$, while $D1$ remained the most challenging class at $\approx 83.2\%$. This contrast underscores that CNN—despite an overall accuracy of 96.38%—is suited to snapshot detection and weakens when temporal gas dynamics must be captured, which explains LSTM's superiority in progressive prediction. The results further show that exploiting the temporal evolution of CO_2 and H_2O was decisive for boosting accuracy in Normal/ $T1$ / $T2$, consistent with their physical roles in tracking thermal aging and moisture and their impact on insulation. Consequently, incorporating CO_2 and H_2O did not only improve numerical accuracy; it also added explanatory value by enabling the model to represent the underlying oil–paper degradation mechanisms in an integrated manner—an advantage over studies that used only five or six traditional gases and overlooked long-term aging and moisture-related degradation, two of the most common root causes of field failures.

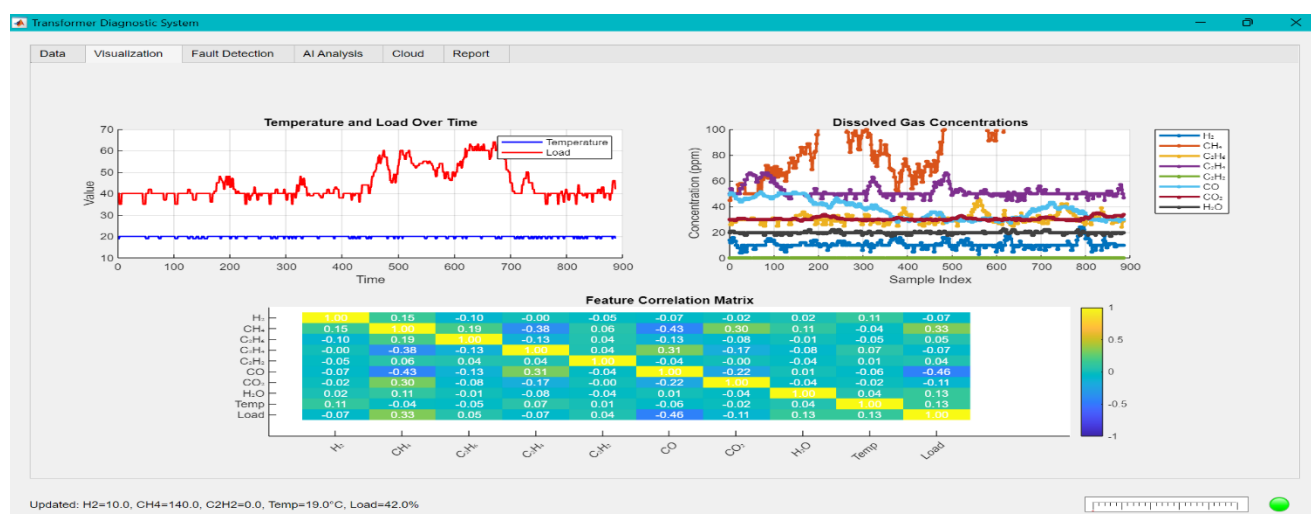


Figure. 9 Correlation Heatmap of Key DGA data (feature correlation matrix)

Fig. 9 the correlation Heatmap of gases (feature correlation matrix) provides an interpretive lens that supports these quantitative findings. Correlations among most gases are weak to moderate, implying that each gas supplies complementary, non-redundant information, which is crucial for successful sequence learning. Particularly salient relationships include $\text{CH}_4\text{--C}_2\text{H}_2 = -0.38$, which naturally separates the thermal pathway (higher CH_4) from the discharge pathway (higher C_2H_2); $\text{CH}_4\text{--CO} = -0.43$, indicating that low-temperature thermal markers typically do not coincide with elevated CO from cellulose degradation, facilitating the discrimination of T1 from paper-insulation issues; and $\text{CO--CO}_2 = -0.22$, which suggests dominance of a single pathway in many cases (either CO or CO_2), making the CO/CO_2 ratio a useful diagnostic indicator as a forward-looking recommendation. In addition, $\text{CO--H}_2\text{O} = -0.46$ shows that moisture increases generally do not coincide with CO spikes at the same instant, supporting the separation of moisture-driven conditions from direct paper degradation. On the positive side, the mild links $\text{H}_2\text{--CH}_4 = +0.15$ and $\text{H}_2\text{--H}_2\text{O} = +0.11$, together with $\text{CH}_4\text{--H}_2\text{O} = +0.33$, confirm that H_2 is a broadly present gas across multiple discharge types and that moisture can co-occur with early thermal activity; LSTM captures this time-dependent interplay, improving early separation of T1/T2.

D1—being a low-energy discharge—remains the most ambiguous due to its transitional nature. When C_2H_2 appears at borderline/intermediate levels while $\text{CO}_2/\text{H}_2\text{O}$ are weak, a sample tends to resemble early PD to the classifier; conversely, when rising C_2H_2 coincides with H_2 , the decision may be pushed toward D2 as a higher-energy arc. These behaviors explain why D1 recall remains $\approx 85.2\%$ despite near-perfect performance elsewhere and motivate the need for more precise temporal and ratio-based representations of gases that characterize this boundary region.

From an industrial standpoint, the impact of this framework is amplified when integrated into a cloud-based Transformer Diagnostic System (TDS) that links periodic DGA measurements with real-time operational variables (temperature, pressure, and load) and issues actionable alerts before hazardous levels are reached. The system thus goes beyond storing or visualizing measurements and provides a predictive function that bridges academic research with field practice, supporting proactive maintenance decisions and reducing service outages and costs.

In light of the foregoing, a set of methodological recommendations emerges, consistent with the LSTM-based results and with the reading of Fig. 9. It

is beneficial to enrich the model inputs with ratio- and time-derived features—such as $\text{C}_2\text{H}_2/\text{H}_2$ and CO/CO_2 , together with $\Delta\text{C}_2\text{H}_2/\Delta t$ and $\Delta\text{H}_2\text{O}/\Delta t$ —reflecting the physical separations indicated by the negative correlations of -0.38 ($\text{CH}_4\text{--C}_2\text{H}_2$) and -0.46 ($\text{CO--H}_2\text{O}$), thereby strengthening the separation of D1 from PD/D2 in transitional cases. It is also advisable to study the effect of the temporal-window length in training and evaluation specifically on D1 recall, verify generalization across assets and operating environments, and calibrate early-warning thresholds to depend on rates of change rather than absolute levels alone. Taken together with operational integration within TDS, these steps transform the proposed framework from a precise diagnostic tool into a comprehensive early-warning and predictive-maintenance system capable of improving transformer reliability and mitigating the economic impact of unexpected failures.

The LSTM architecture was adopted as the primary proposed model, after which it was compared with three other sequential models—GRU, TCN, and Informer—using the same 3,000-sample DGA dataset. As shown in Fig. 7, the LSTM achieved the highest performance, with an accuracy of 97% and an average F1-score of 0.969, compared to 93% / 0.919 for GRU, 91% / 0.899 for TCN, and 92% / 0.909 for the Informer model. Although the Holm–Bonferroni statistical test did not indicate significant differences at $\alpha = 0.05$ based solely on F1-scores, the boxplots in Fig. 7 clearly demonstrate the superior stability and consistently high performance of LSTM, supporting its selection as the core architecture of the proposed framework.

Since the primary goal of this study is to shift from instantaneous diagnosis to real early-fault prediction, an advanced time-based analysis was performed under the LOAO scheme, as illustrated in Fig. 8. With a hysteresis threshold of 0.65 and a cooldown period of 6 hours, all models showed comparable capability in generating early warnings, achieving an average Lead-Time of approximately 8.3 hours before the actual fault and an average TTD of about 1 hour after fault onset. However, the LSTM and GRU exhibited a longer Expected Warning Horizon ($\text{EWH} \approx 3.1 \pm 0.9$ hours) compared to TCN and Informer, which produced significantly shorter warning windows ($\approx 2.1 \pm 0.2$ hours), while maintaining similar False-Alarm Rates of

around 0.38 versus 0.35 alarms/hour. The statistical results in Fig. (Y) indicate that the differences between LSTM and both TCN and Informer are significant ($p < 0.05$), confirming that LSTM not only excels in static F1-score performance but also delivers a more practical predictive capability that provides maintenance teams with a longer intervention window before the fault escalates.

These findings clearly demonstrate that the proposed LSTM-based framework successfully achieves the core objective of this research: accurate early-fault prediction, rather than merely identifying the current operating state. This makes it more suitable for real-world operational deployment compared to traditional methods or alternative models.

10. Conclusion

The findings of this study provide important insights into the advantages of employing sequential analysis for transformer fault diagnosis based on Dissolved Gas Analysis (DGA). Compared with conventional ratio- or threshold-based techniques (e.g., IEC ratios and Duval Triangle), which generally achieve 70–80% accuracy, the proposed LSTM-based framework demonstrated a significant improvement, achieving 99% overall accuracy. This improvement can be attributed to its ability to capture the temporal evolution of gases, particularly gradual increases in H_2 and C_2H_2 , which are key indicators of developing faults. When compared with previous research, notable differences emerge. For instance, study [19] (2020) applied a hybrid CNN + LSTM + DS model using 528 samples covering five gases. While it reported 96.9% accuracy, the dataset was static and relatively small, with time-series characteristics simulated through artificial sliding windows. Similarly, study [44] (2024) adopted TimesNet + Informer on 260 public samples and 350 validation records, transforming 1D signals into 2D via FFT. Although this approach achieved 96.1% accuracy, its performance remained constrained by limited dataset size and coverage. In contrast, the present study leveraged a larger, field-collected dataset of 3,000 sequential samples from Iraqi power transformers, including all eight gases (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , CO , CO_2 , H_2O) alongside operational parameters (temperature, load, pressure). This richer dataset enabled better representation of fault progression and improved generalization. An additional advantage of the proposed framework lies in its cloud integration (ThingSpeak + MongoDB

Atlas), which enabled real-time data acquisition, storage, and remote accessibility. This feature bridges the gap between academic research and industrial deployment, supporting practical predictive maintenance strategies. Moreover, benchmarking against standardized datasets such as IEC TC10 and KEPCO confirmed the robustness of the proposed system, as the LSTM model aligned with the actual inspection-based conditions in most cases. This validation highlights the system's adaptability beyond local field data. The ability to not only detect faults but also predict their progression underscores the practical relevance of this research. Unlike traditional methods that provide a static snapshot of transformer condition, the proposed system offers dynamic early warning by learning sequential gas patterns. This distinction between fault detection (current state) and fault prediction (future risk) is particularly important in power system reliability, where proactive maintenance can significantly reduce unexpected outages and economic losses. In summary, the discussion highlights that while earlier studies such as [19] and [44] provided valuable contributions—emphasizing hybrid models and time-series perspectives—they were constrained by dataset limitations and lack of field-scale validation. By contrast, the present work combines a comprehensive dataset, advanced LSTM modelling, and cloud-enabled monitoring to achieve superior accuracy and practical applicability, making it a strong candidate for deployment in modern smart grid environments.

- Symbol and variables in equations.

A. Pearson Correlation - Eq. (1)

Symbol	Description
x_i	Individual observation of variable X
(y_i)	Individual observation of variable Y
$\bar{x}$	Mean value of X
$\bar{y}$	Mean value of Y
(n)	Total number of samples
(r)	Pearson correlation coefficient

B. Early-Warning Labeling - Eq. (2)

Symbol	Description
(x_t)	Input feature vector at time step t
$x_t \rightarrow t+5$	Sliding window sequence (length = 6)
$(y_{\{t+1\}})$	Target label for next time step
(t)	Current time index

C. LSTM Gate Equations - Eqs. (3) to (8)

Symbol	Description
(f_t)	Forget gate activation
(i_t)	Input gate activation
(o_t)	Output gate activation
(C_t)	Cell state at time t
(h_t)	Hidden state at time t
$\bar{C}_t$	Candidate cell state
$\sigma(\cdot)$	Sigmoid activation function
$\tanh(\cdot)$	Hyperbolic tangent activation
(W_f, W_i, W_o, W_c)	Weight matrices of LSTM gates
(b_f, b_i, b_o, b_c)	Bias vectors of LSTM gates

D. Optimal Threshold Equation - Eq. (9)

Symbol	Description
τ	Optimal decision threshold
$y^{\wedge}$	Predicted output probability
(k)	Required consecutive steps in hysteresis rule
Hysteresis	Rule to reduce false alarms
Cooldown	Waiting time after each triggered alert

E. Training and Evaluation Terms Linked to the Mathematical Formulas

Symbol	Description
Softmax	Multiclass output activation function
Loss	Categorical cross-entropy loss function
Adam	Optimizer used for sequence learning
Batch Size	Number of samples per batch
Epoch	Number of training iterations
Accuracy	Classification accuracy metric
F1	F1-Score
MCC	Matthews Correlation Coefficient

Conflicts of Interest

The authors declare no conflict of interest

Author Contributions

Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, and visualization were carried out by the first author, Ayad Alsobhani. Supervision and project administration were

conducted by the second and third authors, Shamam Alwash and Laith Ali Abdul Rahaim.

References

- [1] X. Wu, Y. He, J. Duan, "A deep parallel diagnostic method for transformer dissolved gas analysis", *Applied Sciences*, Vol. 10, No. 4, 2020, doi: 10.3390/app10041329.
- [2] J. C. Fernández, L. B. Corrales, F. H. Hernández, I. F. Benítez, J. R. Núñez, "A fuzzy logic proposal for diagnosis multiple incipient faults in a power transformer", *Progress in Artificial Intelligence and Pattern Recognition*, pp. 187-198, 2021.
- [3] A. Hechifa, A. Lakehal, A. Nanfak, L. Saidi, C. Labiod, "Machine learning algorithms fusion based on DGA data for improving fault diagnosis of electrical power transformer", *Scientific Bulletin of Electrical Engineering Faculty*, Vol. 23, No. 2, pp. 5-11, 2023, doi: 10.2478/sbeef-2023-0014.
- [4] M. Beheshti Asl, I. Fofana, F. Meghnefi, "Review of various sensor technologies in monitoring the condition of power transformers", *Energies*, Vol. 17, No. 14, 2024, doi: 10.3390/en17143533.
- [5] S. S. M. Ghoneim, M. Baz, A. Alzaed, Y. T. Zewdie, "Predicting the insulating paper state of the power transformer based on XGBoost/LightGBM models", *Scientific Reports*, Vol. 15, No. 1, Art. 17836, 2025, doi: 10.1038/s41598-025-03033-4.
- [6] M. Badawi, et al., "Reliable estimation for health index of transformer oil based on novel combined predictive maintenance techniques", *IEEE Access*, Vol. 10, pp. 25954-25972, 2022, doi: 10.1109/access.2022.3156102.
- [7] Z. Wang, Q. Liu, X. Wang, P. Jarman, G. Wilson, "Discussion on possible additions to IEC 60897 and IEC 61294 for insulating liquid tests", *IET Electric Power Applications*, Vol. 5, No. 6, pp. 486-493, 2011.
- [8] K. Jongvilaikasem, N. Pattanadech, "Dissolved gas behavior in natural ester under corona, surface, and internal discharges", In: *Proc. of 8th Int. Conf. Condition Monitoring and Diagnosis (CMD)*, pp. 326-329, 2020.
- [9] C. Rajotte, "Maintenance of transformers and reactors", *Transformer and Reactor Life Management*, pp. 229-251, 2024.
- [10] R. R. Rogers, "IEEE and IEC codes to interpret incipient faults in transformers, using gas in oil analysis", *IEEE Transactions on Electrical Insulation*, No. 5, pp. 349-354, 2007.

- [11] M. Duval, "A review of faults detectable by gas-in-oil analysis in transformers", *IEEE Electrical Insulation Magazine*, Vol. 18, No. 3, pp. 8-17, 2002.
- [12] G. Odongo, R. Musabe, D. Hanyurwimfura, "A multinomial DGA classifier for incipient fault detection in oil-impregnated power transformers", *Algorithms*, Vol. 14, No. 4, Art. 128, 2021.
- [13] M. J. Lin, "A new approach with three dimension figure and ANSI/IEEE C57.104 standard rule diagnoses transformer's insulating oil", *Engineering*, Vol. 6, No. 12, pp. 841-848, 2014.
- [14] Z. Wang, I. Cotton, S. Northcote, "Dissolved gas analysis of alternative fluids for power transformers", *IEEE Electrical Insulation Magazine*, Vol. 23, No. 5, pp. 5-14, 2007.
- [15] International Electrotechnical Commission, *IEC 62770: Fluids for Electrotechnical Applications - Unused Natural Esters for Transformers and Similar Electrical Equipment*, 2013.
- [16] U. M. Rao, I. Fofana, J. S. N'cho, "On some imperative IEEE standards for usage of natural ester liquids in transformers", *IEEE Access*, Vol. 8, pp. 145446-145456, 2020.
- [17] U. M. Rao, Y. Sood, R. Jarial, "Review on ester based dielectric liquids for transformer insulation technology", In: *Proc. of Int. Conf. Condition Assessment Techniques in Electrical Systems (CATCON)*, pp. 1-6, 2015.
- [18] T. Rouse, "Mineral insulating oil in transformers", *IEEE Electrical Insulation Magazine*, Vol. 14, No. 3, pp. 6-16, 2002.
- [19] X. Wu, Y. He, J. Duan, "A deep parallel diagnostic method for transformer dissolved gas analysis", *Applied Sciences*, Vol. 10, No. 4, 2020, doi: 10.3390/app10041329.
- [20] O. M. Elmabrouk, F. A. Masoud, N. S. Abdelwanis, "Diagnosis of power transformer faults using fuzzy logic techniques based on IEC ratio method", In: *Proc. of 6th Int. Conf. Engineering & MIS*, pp. 1-5, 2020.
- [21] X. Yang, W. Chen, A. Li, C. Yang, "A hybrid machine-learning method for oil-immersed power transformer fault diagnosis", *IEEE Transactions on Electrical and Electronic Engineering*, Vol. 15, No. 4, pp. 501-507, 2020, doi: 10.1002/tee.23081.
- [22] J. C. Fernández, L. B. Corrales, F. H. Hernández, I. F. Benítez, J. R. Núñez, "A fuzzy logic proposal for diagnosis multiple incipient faults in a power transformer", In: *Proc. of Progress in Artificial Intelligence and Pattern Recognition*, pp. 187-198, 2021.
- [23] G. Odongo, R. Musabe, D. Hanyurwimfura, "A multinomial DGA classifier for incipient fault detection in oil-impregnated power transformers", *Algorithms*, Vol. 14, No. 4, Art. 128, 2021, doi: 10.3390/a14040128.
- [24] I. B. M. Taha, S. Ibrahim, D. E. A. Mansour, "Power transformer fault diagnosis based on DGA using a convolutional neural network with noise in measurements", *IEEE Access*, Vol. 9, pp. 111162-111170, 2021, doi: 10.1109/access.2021.3102415.
- [25] M. Badawi, et al., "Reliable estimation for health index of transformer oil based on novel combined predictive maintenance techniques", *IEEE Access*, Vol. 10, pp. 25954-25972, 2022, doi: 10.1109/access.2022.3156102.
- [26] S. A. E. S. Ward, S. A. Ibrahim, A. E. Faraskoury, D.-E. A. Mansour, M. Badawi, "Identification of transformer oil incipient faults based on the integration between different DGA techniques", *Delta University Science Journal*, Vol. 6, No. 1, pp. 412-421, 2023.
- [27] V. Rokani, S. D. Kaminaris, P. Karaisas, D. Kaminaris, "Power transformer fault diagnosis using neural network optimization techniques", *Mathematics*, Vol. 11, No. 22, Art. 4693, 2023, doi: 10.3390/math11224693.
- [28] S. Guan, H. Yang, T. Wu, "Transformer fault diagnosis method based on TLR-ADASYN balanced dataset", *Scientific Reports*, Vol. 13, No. 1, Art. 23010, 2023, doi: 10.1038/s41598-023-49901-9.
- [29] A. Hechifa, A. Lakehal, A. Nanfak, L. Saidi, C. Labiod, "Machine learning algorithms fusion based on DGA data for improving fault diagnosis of electrical power transformer", *Scientific Bulletin of Electrical Engineering Faculty*, Vol. 23, No. 2, pp. 5-11, 2023, doi: 10.2478/sbeef-2023-0014.
- [30] Y. Wang, "Application of improved support vector machine model in fault diagnosis and prediction of power transformers", *Advanced Control Applications: Engineering and Industrial Systems*, Vol. 6, No. 4, Art. e170, 2023, doi: 10.1002/adc2.170.
- [31] X. Miao, et al., "Fault diagnosis of oil-immersed transformers based on the improved neighborhood rough set and deep belief network", *Electronics*, Vol. 13, No. 1, Art. 5, 2023, doi: 10.3390/electronics13010005.
- [32] M. K. Ngwenyama, M. N. Gitau, "Discernment of transformer oil stray gassing anomalies using machine learning classification techniques", *Scientific Reports*, Vol. 14, No. 1, Art. 376, 2024, doi: 10.1038/s41598-023-50833-7.

- [33] C. Luo, et al., "Diffusion characteristics of dissolved gases in oil under different oil flow circulations", *Energies*, Vol. 18, No. 2, Art. 432, 2025, doi: 10.3390/en18020432.
- [34] M. F. Naveed, A. Amaar, S. S. Khan, M. Omar, S. Larkin, "Detection and continuous monitoring of moisture content in transformer oil using fractal-based capacitive sensor", *Heliyon*, Vol. 10, No. 24, Art. e40995, 2024, doi: 10.1016/j.heliyon.2024.e40995.
- [35] A. Aritonang, S. Prayogi, "Analysis of GT 2.1 transformer oil testing using the dissolved gas analysis (DGA) method at PT PLN Nusantara Power", *G-Tech: Journal Teknologi Terapan*, Vol. 9, No. 1, pp. 540-550, 2025, doi: 10.70609/gtech.v9i1.6481.
- [36] S. Nisworo, A. Hasibuan, M. Isa, I. M. A. Nrratha, M. Jannah, M. Muhammad, "Assessing transformer health through analysis of dissolved gases in cooling oil", *International Journal of Applied Power Engineering (IJAPE)*, Vol. 13, No. 4, pp. 1014-1020, 2024, doi: 10.11591/ijape.v13.i4.pp1014-1020.
- [37] L. Jin, D. Kim, K. Y. Chan, A. A. Siada, "Deep machine learning-based asset management approach for oil-immersed power transformers using dissolved gas analysis", *IEEE Access*, Vol. 12, pp. 27794-27809, 2024, doi: 10.1109/access.2024.3366905.
- [38] Y. Jia, X. Luo, S. Ji, W. Gao, S. Li, F. Zhang, "Pyrolysis and temperature characteristics of the gas produced by power-frequency arc in transformer oil", *Case Studies in Thermal Engineering*, Vol. 61, Art. 104913, 2024, doi: 10.1016/j.csite.2024.104913.
- [39] S. Bustamante, M. Manana, A. Arroyo, A. Laso, R. Martinez, "Evolution of graphical methods for the identification of insulation faults in oil-immersed power transformers: A review", *Renewable and Sustainable Energy Reviews*, Vol. 199, 2024, doi: 10.1016/j.rser.2024.114473.
- [40] M. Beheshti Asl, I. Fofana, F. Meghnefi, "Review of various sensor technologies in monitoring the condition of power transformers", *Energies*, Vol. 17, No. 14, Art. 3533, 2024, doi: 10.3390/en17143533.
- [41] V. Bargate, A. Yadav, S. Gupta, "Analysis of transformer defect using different dissolved gas analysis techniques: A comparative study", *Journal of The Institution of Engineers (India): Series B*, 2024, doi: 10.1007/s40031-024-01156-2.
- [42] M. Badawi, et al., "On highlighting the merits of gas-to-liquid transformer oil under accelerated thermal aging and faults: Electrical and physio-chemical properties", *IEEE Access*, Vol. 12, pp. 93372-93388, 2024, doi: 10.1109/access.2024.3420230.
- [43] N. Kianimehr, H. Z. Meymand, F. Shahnia, "Classification and fault diagnosis of power transformers with dissolved gas analysis using improved clustering methods", *COMPEL*, Vol. 44, No. 1, pp. 50-66, 2024, doi: 10.1108/compel-06-2024-0250.
- [44] X. Zhang, K. Yang, L. Zheng, "Transformer fault diagnosis method based on TimesNet and Informer", *Actuators*, Vol. 13, No. 2, Art. 74, 2024, doi: 10.3390/act13020074.
- [45] M. K. K. Alabdullh, M. Joorabian, S. G. Seifossadat, M. Saniei, M. Abasi, "A novel method to estimate the lifetime of mineral oil-type power transformers based on the analysis of chemical and physical indicators using artificial intelligence", *Heliyon*, Vol. 10, No. 22, Art. e40447, 2024, doi: 10.1016/j.heliyon.2024.e40447.
- [46] R. Zemouri, "Power transformer prognostics and health management using machine learning: A review and future directions", *Machines*, Vol. 13, No. 2, Art. 125, 2025, doi: 10.3390/machines13020125.
- [47] S. Liu, Z. Xie, Z. Hu, "DGA-based fault diagnosis using self-organizing neural networks with incremental learning", *Electronics*, Vol. 14, No. 3, Art. 424, 2025, doi: 10.3390/electronics14030424.
- [48] M. Duval, A. dePabla, "Interpretation of gas-in-oil analysis using new IEC publication 60599 and IECTC 10 databases", *IEEE Electrical Insulation Magazine*, Vol. 17, No. 2, pp. 31-41, 2001, doi: 10.1109/57.917529.
- [49] S. Kim, S.-H. Jo, W. Kim, J. Park, J. Jeong, Y. Han, D. Kim, B. D. Youn, "A semi-supervised autoencoder with an auxiliary task (SAAT) for power transformer fault diagnosis using dissolved gas analysis", *IEEE Access*, Vol. 8, pp. 178295-178310, 2020, doi: 10.1109/ACCESS.2020.3027830.